



DATA SCIENCE
INSTITUTE

WOLF/

ASSIGNMENT

MACHINE LEARNING II

Email Campaign Classification Analysis

Python

REPORT



Pillar1-e

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BACKGROUND:

The data is a marketing campaign data of a skin care clinic associated with its success.

Description of variables-

Success: Response to marketing campaign of Skin Care Clinic which offers both products and services. (1: email Opened, 0: email not opened)

AGE: Age Group of Customer

Recency_Service: Number of days since last service purchase

Recency_Product: Number of days since last product purchase

Bill_Service: Total bill amount for service in last 3 months

Bill_Product: Total bill amount for products in last 3 months

Gender (1: Male, 2: Female)

Note: Answer following questions using entire data and do not create test data.

QUESTIONS-

1. Import Email Campaign data. Obtain decision tree to classify cases as success=0 or 1. Obtain Sensitivity/Recall using cut-off value as 0.50 for estimated probabilities.
2. Compare performance of Decision Tree and Random Forest Method using area under the ROC curve.
3. Implement Neural Network Algorithm and obtain area under the ROC curve .

ASSIGNMENT SUMMARY OVERVIEW:

This project implements advanced machine learning algorithms including Decision Trees, Random Forests, and Neural Networks to predict email marketing campaign success for a skin care clinic. The analysis compares three distinct modelling approaches using ROC-AUC evaluation, cross-validation, and comprehensive visualization.

Key Features:

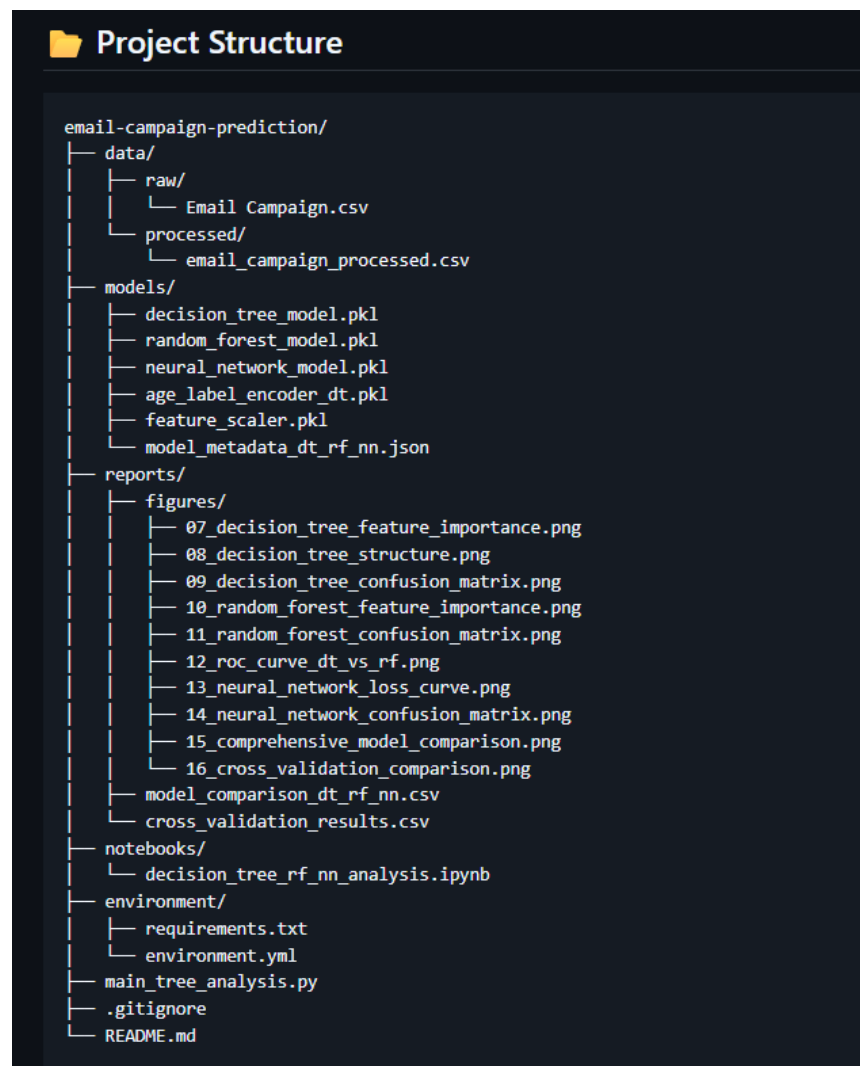
- Decision Tree with pruning and sensitivity analysis at 0.50 threshold
- Random Forest ensemble method with 100 trees
- Neural Network (MLP) with adaptive learning and early stopping
- ROC-AUC Comparison across all models
- Cross-Validation (5-Fold Stratified) for realistic performance estimates
- Feature Importance Analysis for interpretability
- Comprehensive Visualizations (10 plots including confusion matrices, ROC curves, radar charts)

Dataset: 683 customer records with demographics, purchase recency, billing history, and email response data.

Note: As per assignment requirements, models are trained on the entire dataset without train/test split. Cross-validation results are provided for more realistic performance estimates.

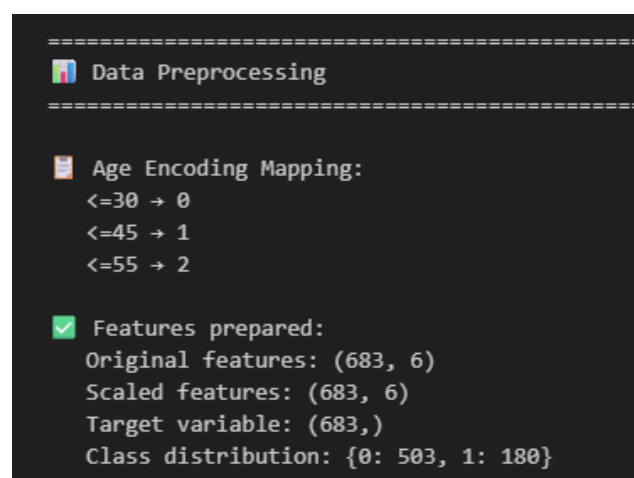
Seeing as we are not doing a train/test split on the data the AUC scores can be prone to being evaluated on an artifact or can be easily overfit as the evaluation would be seeing the same data from training/modelling during analysis. We standardised the features which was important for doing the neural network. We also limited the depth of the decision tree to prevent overfitting, and we reserved a 10% validation fraction for our neural network to use as validation data to also prevent overfitting.

PROJECT STRUCTURE:

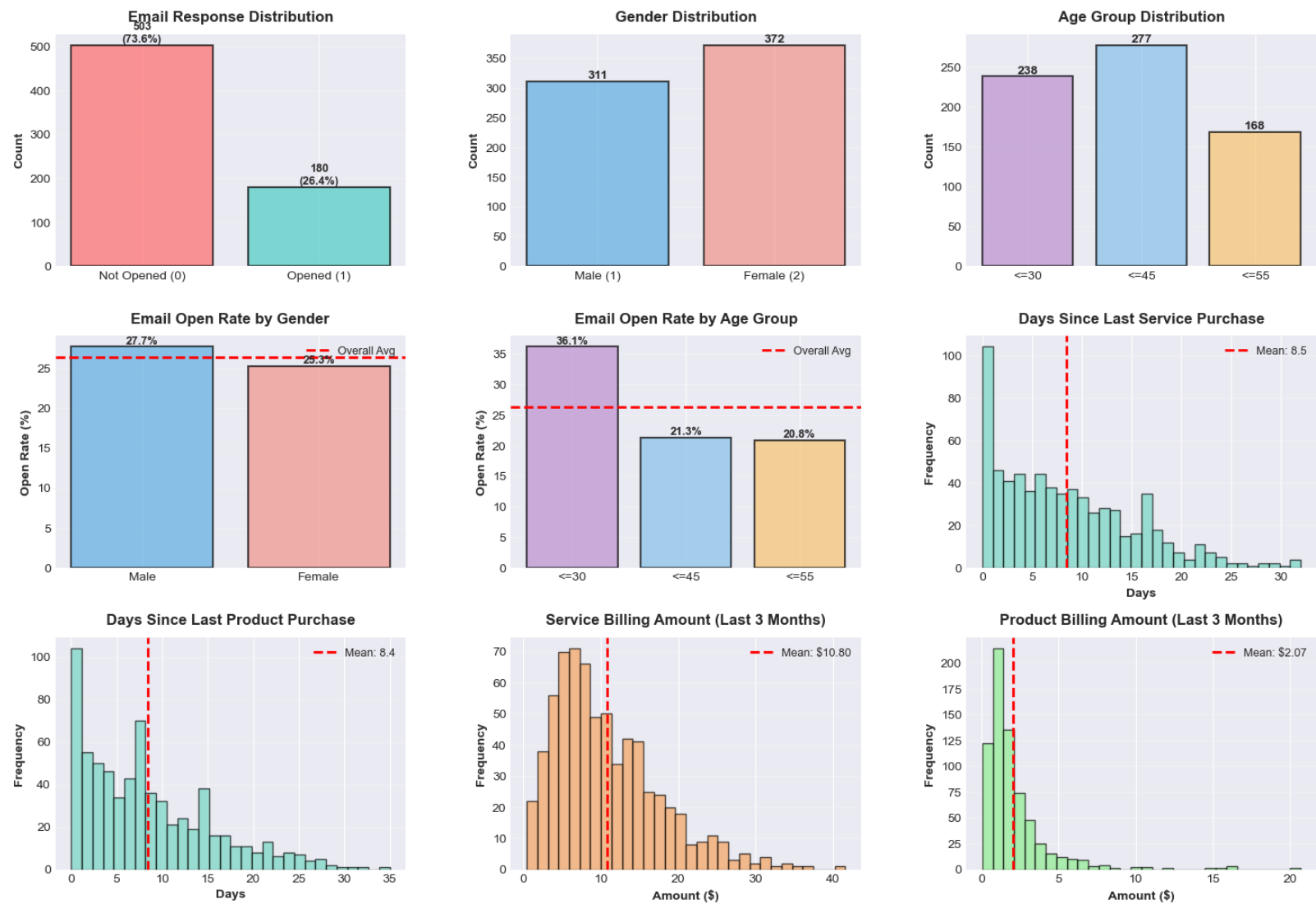


Data Pre-Processing:

Data was loaded successfully and prepared for analysis, age encoding was



Email Campaign - Exploratory Data Analysis Dashboard



QUESTION 1: Import Email Campaign data. Obtain decision tree to classify cases as success=0 or 1. Obtain Sensitivity/Recall using cut-off value as 0.50 for estimated probabilities.

Decision Tree Classification

Tree depth: 8

Number of leaves: 42

Number of features: 6

DECISION TREE PERFORMANCE (Cutoff = 0.50)

Classification Report:					
		precision	recall	f1-score	support
Not Opened	Not Opened	0.8815	0.9165	0.8986	503
	Opened	0.7375	0.6556	0.6941	180
accuracy				0.8477	683
macro avg		0.8095	0.7860	0.7964	683
weighted avg		0.8435	0.8477	0.8447	683

SENSITIVITY/RECALL (Success=1) at cutoff 0.50: 0.6556 (65.56%)

→ Out of all actual email openers, model correctly identifies **65.56%**

Additional Metrics:

Accuracy: 0.8477 (84.77%)
Precision: 0.7375 (73.75%)
F1-Score: 0.6941

Feature Importance:

Bill_Service: 0.400637
Recency_Service: 0.226562
Bill_Product: 0.197016
Recency_Product: 0.146029
AGE_Encoded: 0.018455
Gender: 0.011301

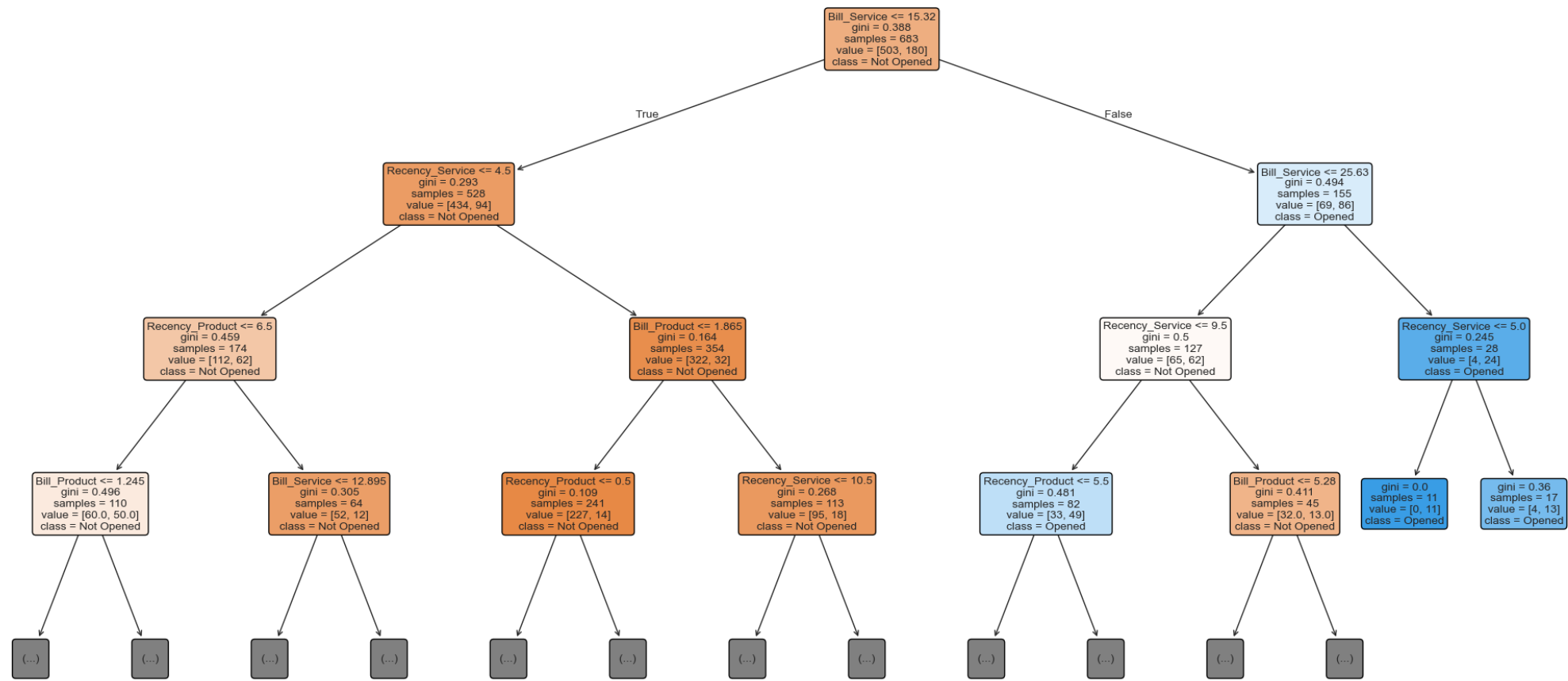
Additional Metrics:

Accuracy: 0.8477 (84.77%)
Precision: 0.7375 (73.75%)
F1-Score: 0.6941



Visualise Decision Tree

Decision Tree Structure (Top 3 Levels)



QUESTION 2: Compare performance of Decision Tree and Random Forest Method using area under the ROC curve.

Random Forest Classification

Number of trees: 100

Max tree depth: 10

Number of features: 6

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RANDOM FOREST PERFORMANCE
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Classification Report:

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		precision	recall	f1-score	support
Not	Opened	0.8513	0.9563	0.9007	503
	Opened	0.8136	0.5333	0.6443	180
accuracy				0.8448	683
macro avg		0.8324	0.7448	0.7725	683
weighted avg		0.8414	0.8448	0.8332	683

Performance Metrics:

Accuracy: 0.8448 (84.48%)

Precision: 0.8136 (81.36%)

Recall: 0.5333 (53.33%)

F1-Score: 0.6443

Random Forest Feature Importance:

Bill_Service: 0.367358

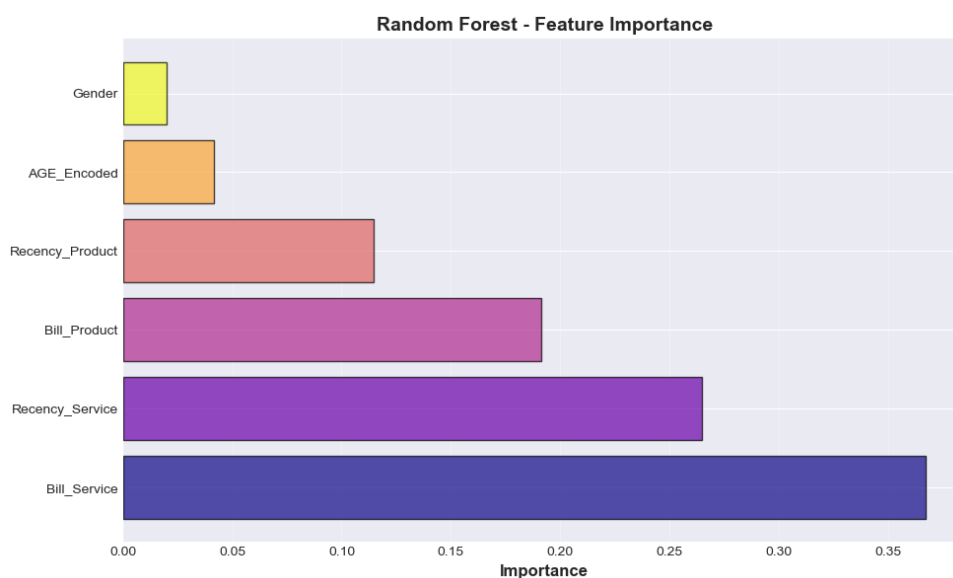
Recency_Service: 0.264759

Bill_Product: 0.191347

Recency_Product: 0.114864

AGE_Encoded: 0.041654

Gender: 0.020018



ROC Curve Comparison - Decision Tree vs Random Forest

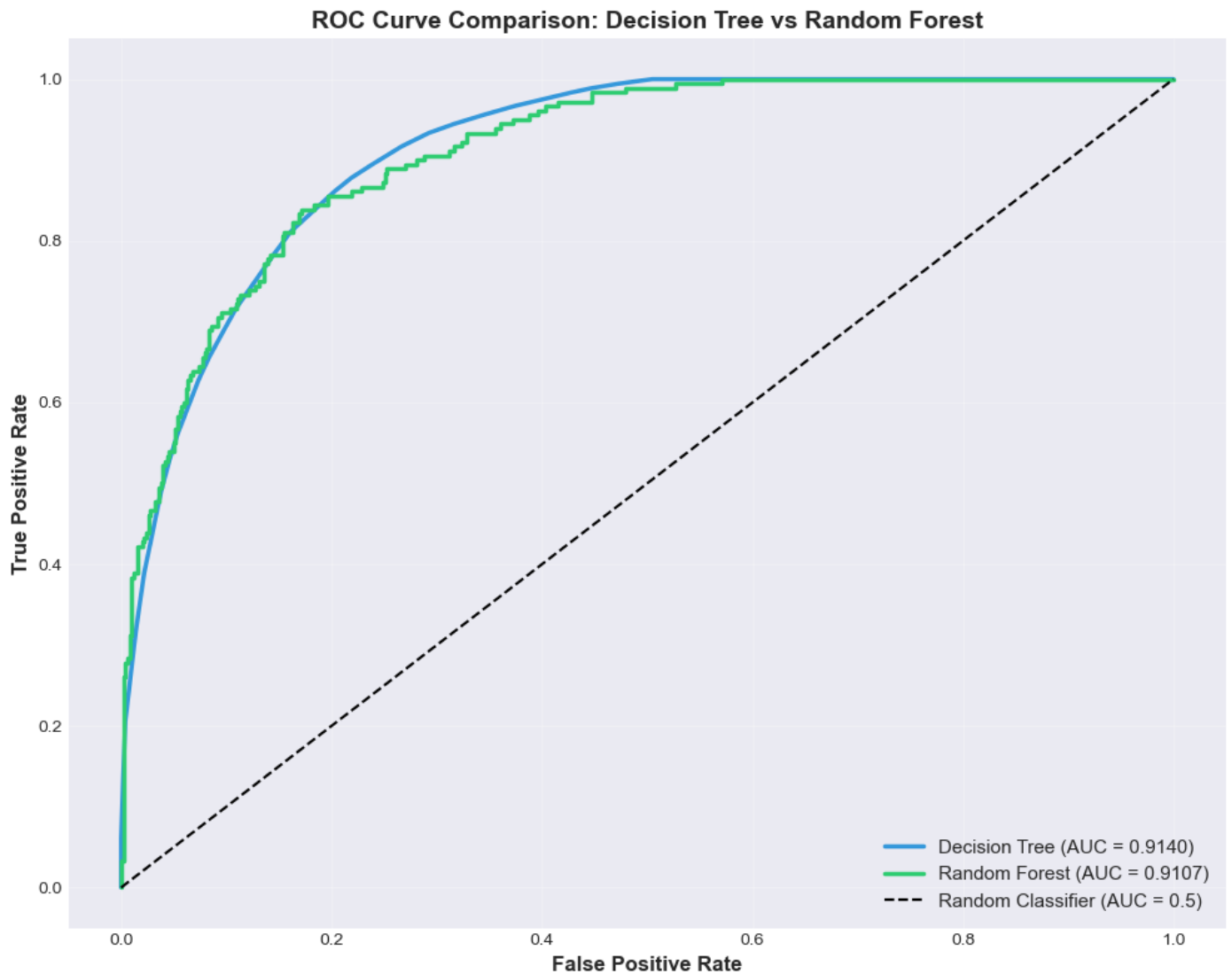
Area Under ROC Curve (AUC):

Decision Tree: 0.9140

Random Forest: 0.9107

Difference: 0.0033

Winner: Decision Tree (better by **0.33%**)



QUESTION 3: Implement Neural Network Algorithm and obtain are under the ROC curve .

Neural Network Classification

Architecture: Input → Hidden(100) → Hidden(50) → Output

Neural Network trained successfully!

Number of iterations: 12

Number of layers: 4

Final loss: 0.426709

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NEURAL NETWORK PERFORMANCE
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📊 Classification Report:

      precision    recall  f1-score   support

Not Opened      0.8167      0.8946      0.8539        503
   Opened       0.5985      0.4389      0.5064        180

   accuracy              0.7745        683
  macro avg       0.7076      0.6668      0.6802        683
 weighted avg     0.7592      0.7745      0.7623        683
...

📈 Area Under ROC Curve (AUC): 0.7634
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🏆 COMPREHENSIVE MODEL COMPARISON
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📊 PERFORMANCE SUMMARY TABLE:
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Model	Accuracy	Precision	Recall	F1-Score	AUC	Rank
Decision Tree	0.847731	0.737500	0.655556	0.694118	0.914032	1
Random Forest	0.844802	0.813559	0.533333	0.644295	0.910736	2
Neural Network	0.774524	0.598485	0.438889	0.506410	0.763408	3

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🏆 BEST PERFORMING MODEL
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Model:      Decision Tree
AUC:        0.9140
Accuracy:    0.8477
Precision:   0.7375
Recall:      0.6556
F1-Score:    0.6941
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Visualisation of Comprehensive Model Comparison:

